

Network Optimization Summary for NUS

Goal

The goal is to find the best network to service the same demand for the lowest cost while maintaining a similar transit time. \

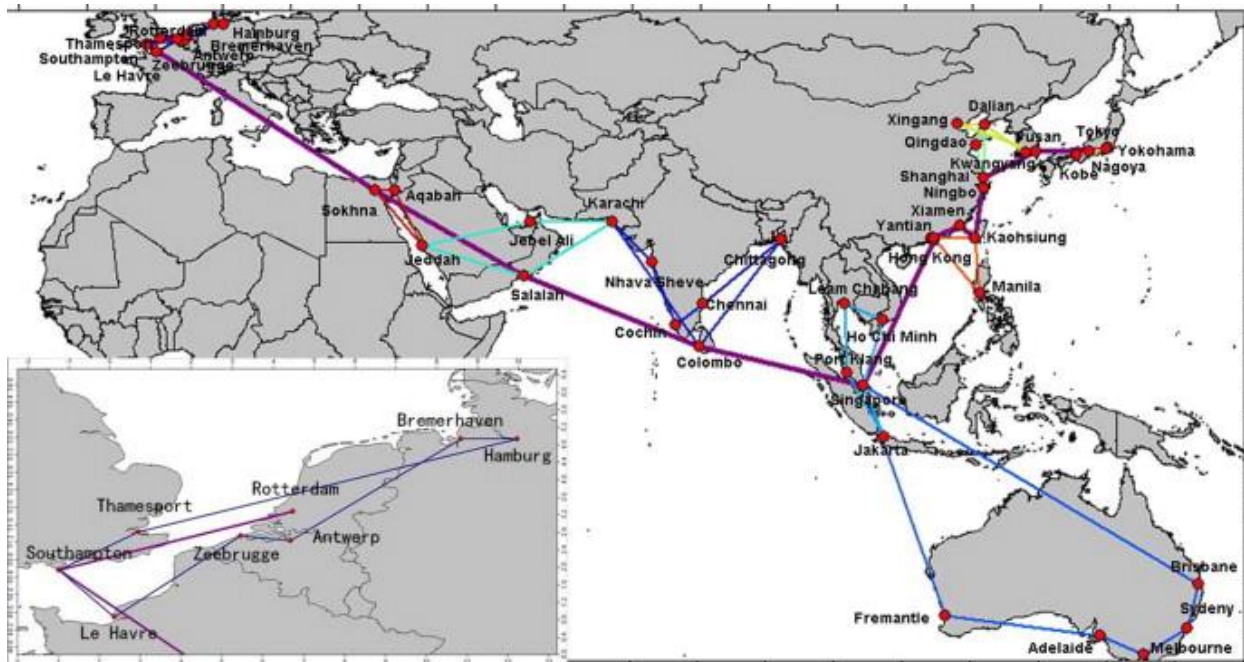
The idea is to take all the demand between two ports (Shanghai to Los Angeles, Bangkok to Rotterdam, ...) and see what would be the best lines that would serve this demand. Of course, the best network is useless if the routing on it is badly done so this project requires 2 parts.

- Network creation
- Routing optimization

Both will iteratively improve each other.

This is an example of our problem.

- There is a purple line going from China to Europe
 - This is the big CMA line that exports the goods from Asia to Europe
- There are a lot of other smaller lines.
 - For instance, the light blue between Vietnam, Thailand, Singapore
 - The goals of these lines are to gather the export from small Asian countries and to send it to Singapore where it will be transshipped on the big lines to go to Europe.
 - They also help the intra-Asia market (Indonesia for instances buy rice from Vietnam, Korea buys banana from the Philippines, ...)
- The big lines between continents are controlled in Marseille (CMA is a French company) but the small lines in Asia are controlled here in Singapore.
 - The goal of this project is to find the best network made of all the small lines that would serve the same demand.
 - In a reasonable time
 - For an optimized cost
 - There are two ways to approach it
 - Take our current network and optimize it
 - Create a network from scratch.
 - Both options are interesting to us



Prototype

The prototype was built with these constraints

Objective

Minimize network costs by providing the best lines that:

- Reduce chartering cost
- Reduce bunkering cost
- Reduce transshipment cost (moves + storage)
- Reduce port call cost

Constraints

- New transit days < existing transit days + 3 (we allow some flexibility in the routing to see potential optimisation)
- Frequency of new lines designed kept at 7 days (a vessel should call a port every week on the same day)
- Vessel draft < max port draft (some vessels are too big to enter some small Asian ports)

Output expected: Optimal line design detailing the port, terminals, and vessels to be deployed.

Remarks

- Demand can flow on all edges (no constraint)
 - o For instance,

- Cargo transshipment ports can be changed
 - Cargo can take different lines from original
- Goal is to meet 100% of the demand, but this constraint can be relaxed if it takes too long to develop

The first phase was made as a POC, meaning there were some simplifications made. For the research projects, it will be possible to go deeper and to make the project more realistic.

Simplifications

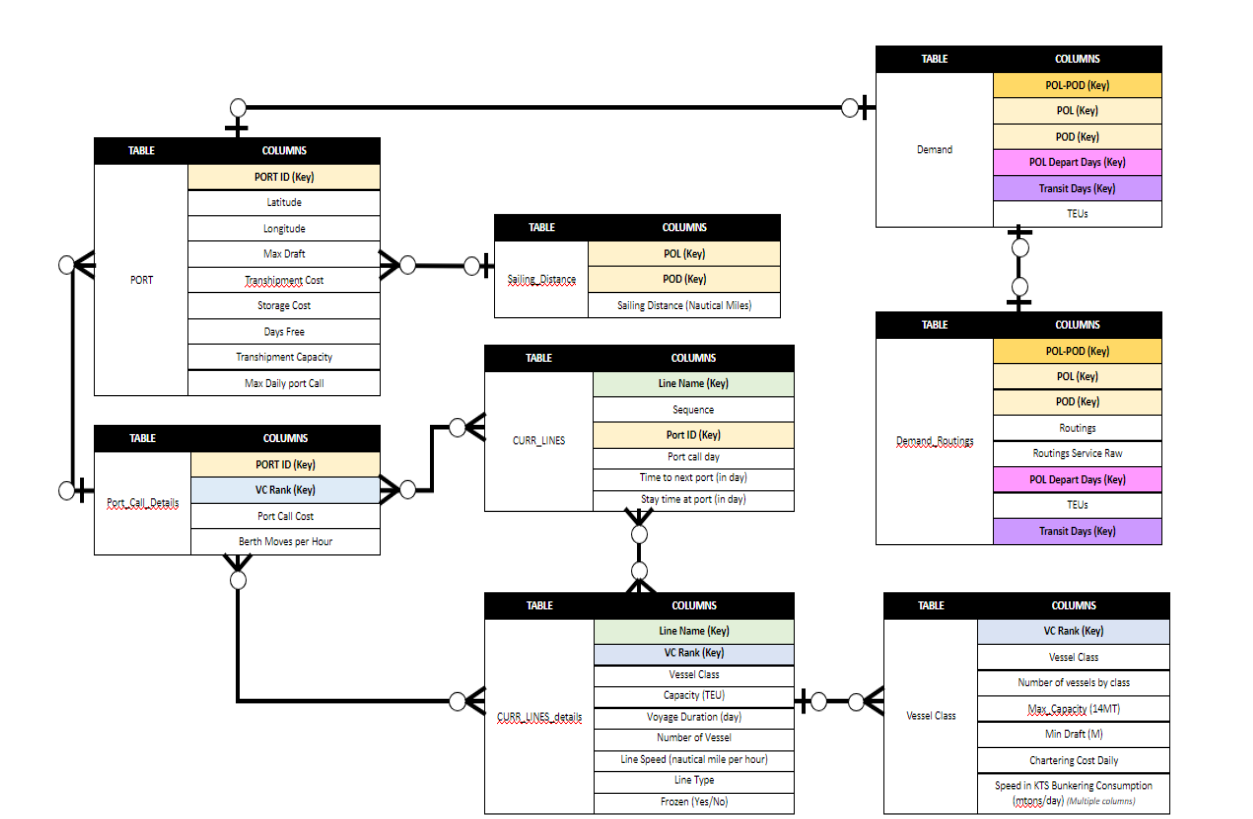
- The following elements are not considered.
 - The empty containers
 - It means that some lines might be over-optimized by the algorithm where in reality some space is required for empties.
 - Terminal incentive
 - 20/40 split, reefer (vessel plug), special cargo,
 - everything at TEU level for the moment
- The following elements are simplified.
 - Port level instead of terminal level
 - Vessel class instead of vessel level

Input

This is the diagram of all the input data.

The two most important to understand are

- demand
- lines



Demand

- A weekly average volume exported from / imported into APAC region from Jul 2023 - Sep 2023. We define it as an APAC shipment as falling within this period if
 - POL or POD falls within ASIA or OCEANIA
 - the **arrival date** of the shipment at POD is between these dates.

Lines

For this study, services under all brands (ANL, APL, CMA, CNC) have been considered. As we are the APAC region, what we can really control (in terms of pricing and line design) are the services under the CNC brand - which serves the intra-asia region, and 2 specific lines in ANL. The other lines can be used for flowing the cargo but cannot be modified.

The lines were taken in Early October.

- to compute historical network cost:
 - bunkering cost (depending on length of service route and speed of service)
 - chartering cost (depending on class of vessel deployed)
 - port call cost (depending on class of vessel deployed)

Output

The output will be

- Network made of
 - The CMA lines that are not touched (but used for routing)
 - The created lines in Asia
- Demand routing
 - How is the demand is routed on this new network
 - With the transit time
- KPI
 - Cost to be able to compare the historical and the created networks.
 - Line utilization